change that would alter the total charge present in the system. Hence, the change of a proton into a neutron, or of a neutron into a proton, cannot take place in the nucleus without compensation. Otherwise, many atoms which are actually considered stable, would have to be unstable. There is, for instance, a boron nucleus having the (not exact) mass number 12, and also a carbon nucleus having the mass number 12. The boron nucleus consists of 7 neutrons and 5 protons, whereas the carbon nucleus has 6 protons and 6 neutrons. Their respective symbols are ${}_5B^{12}$ and ${}_6C^{12}$. The mass of the boron nucleus, however, is slightly greater than that of the carbon nucleus. The difference is 0.013 a.m.u., and consequently, the difference in binding energy is about 12 Mev. The carbon nucleus has a greater mass defect than the boron nucleus, so that its component parts are bound together more tightly than those of the boron nucleus. We can assume, therefore, that the boron nucleus is unstable and changes spontaneously into a carbon nucleus. This transmutation would liberate energy to the extent of 12 Mev. But this process can occur only when a neutron of the boron nucleus changes into a proton. However, by virtue of the law of conservation of electric charge, such a process is possible only if the newly formed elementary quantum of positive electricity is compensated for by the simultaneous formation of an elementary quantum of negative electricity and the latter removed from the nucleus. This could take place through the emission of an electron simultaneously with the change of the neutron into a proton. Actually, this boron nucleus is not a stable structure, but a radioactive one. It emits electrons—in other words, negative beta rays—and changes into a carbon nucleus.

Nevertheless, this transformation would not be possible if the law of the conservation of angular momentum did not hold. As we have mentioned before, the angular momentum, or *spin*, of both a proton and a neutron is $\hbar/2$, to be considered positive or negative according to the spatial orientation of the axis of rotation. Therefore, the angular momentum of a nucleus composed of an even number of particles is always an even multiple of $\hbar/2$. Both the carbon nucleus and the boron nucleus consist of 12 particles; consequently, the nuclear spin of each must be an integer multiple of $\hbar$. But pursuant to the law of the conservation of electric charge, an electron must be emitted during the transmutation process, and this electron, too, has an angular momentum (electronic spin) of $\hbar/2$. Therefore, the resulting carbon nucleus would retain an angular momentum (nuclear spin), the value of which is an odd multiple of $\hbar/2$. In this case, therefore, the angular momenta involved would not balance. But in this predicament we must remember that when we studied natural beta radiation, we found that similar difficulties of balancing the energies furnished us with a clue to the existence of the neutrino, ejected simultaneously with the emission of the electron. The neutrino, obviously,